

AL-2001

Artificial Intelligence

Lab # 3

Objectives:

- List, Dictionary, Tuple, Set

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your logic on Copy / Notebook.
2. Write **Your Name** and **Roll No** on your Paper/Sheet's first page.
3. **Do not copy from any source otherwise you will be penalized with negative marks.**
4. Complete your lab **within given Time Slot**.
5. Paste all your codes along with screenshots in a word file and renamed with your roll number.
6. Keep all your source files in your computer for verification. Do not overwrite a single source file for all programs.

Problems:

1. list = ["apple", "cherry", "orange", "kiwi", "melon", "mango"]
 - a. Remove "cherry" and "melon" from list.
 - b. Add "banana" at second last index.
2. list = [1, 4, 56, 2, 4 , 12, 6, 89 ,11, 0] Write a program to empty list using while loop and pop() function.
3. Write a python code to make a list. Populate it with random numbers using random number generator and then find the min and max value from the list. Size of list should be given by user. Note: do not use max and min built-in function.
4. Write a python code to create a tuple. And do the following tasks.
 - a. Create Tuple = ("john", " mark", 12, "14", " orange", 4.5)
 - b. Then update the Tuple and add a new value 6.5.
 - c. After updating the value, create a function that shows the count of how many integers, strings and float variables does it have.
5. Write a python code to create a dictionary. And do the following tasks.
 - a. Create Dictionary = {"brand": "Samsung", "OS type": "Oreo", "color": "black", "camera": "42 megapixels", "year": 2012}
 - b. Then add a list in the current dictionary with key = "sizes" and values of random numbers.



- c. Then delete the “year” key from the dictionary. iv. Lastly show the dictionary in the following order. Use loop to show the dictionary:
- i. Brand
 - ii. Color
 - iii. Camera
 - iv. OS-type
 - v. Sizes
6. Create a function that takes list list1 and list2 as parameter, create a list list3 inside the function and store the sum of list list1, list2 into list3 then return list3.
- list1 = [11,22,33,44,21,54,67,54,33,222,4]
- list2 = [3,4,5,32,21,33,66,75,87,97,1]
7. Create a function dsort() to sort a list in descending order, taking a list as argument and returning it. Use the following list and print it. list = [5, 6, 7, 23 ,12 ,3, 3, 4 ,5, 12, 34]

You need to do your exercise within the given time.